

Sarvesh Bhattacharyya

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PROFILE SUMMARY

AI & Backend Engineer with production experience building and shipping AI systems independently — from blank slate to live users. Experienced across Agentic and LLM orchestration, RAG pipelines, backend infrastructure. Focused on systems that work under real constraints: latency, cost, deployment, and reliability.

EXPERIENCE

AI Engineer Intern — Beaut Group (business.superowl.in)

Feb 2026 – Present · Bengaluru, Karnataka · On-site

- Integrated a third-party Voice AI platform and independently designed the automation layer for small-business customer interactions, owning backend workflows, webhook orchestration, call-routing logic, and prompt engineering to fine-tune assistant behavior for production-facing agentic interactions.
- Architected concurrent inbound call handling via runtime prompt injection, enabling multiple simultaneous client interactions without creating separate assistant instances or adding instance overhead.
- Built and validated AI-assisted customer interaction workflows across multiple businesses, validating real-world behavior, edge cases, and operational readiness.
- Extended the backend into a multi-tenant owner-facing mobile app with backend services, live call screens, missed-call summaries, push notifications, and real-time sync — delivered end-to-end in under 3 months.

PROJECTS

ROAST — Live-Market Based Resume Critic [GitHub](#) | [Live Demo](#)

FastAPI · RAG · Multi-Agent Pipelines · Redis · WebSocket · SQLite+FTS5 · Groq API · Langfuse · Prompt Engineering · Gemini · React+Vite

- Built and deployed an agentic resume critic that pulls live job-market data before evaluating resumes — 7-agent pipeline (MarketContextAgent, RedFlagAgent, SixSecondAgent, CompetitiveAgent, TechnicalDepthAgent, ReviewAgent, FollowUpAgent) with parallel asyncio execution, WebSocket streaming, Redis session state, and Langfuse observability; 95+ live analyses at \$0/month.
- TechnicalDepth Agent performs real-time DuckDuckGo tool-calling to evaluate projects with genuine technical understanding — not keyword matching; inference chains expose the recruiter's exact thought process per resume weakness.
- Market intelligence store built fully offline: consisted of monthly ingestion pipeline that embeds job-market signals with Gemini embeddings into SQLite FTS5 + sqlite-vec, replacing hosted vector DBs after free-tier suspension. Reduced ingestion time from 10 hrs to 1 hr via rate-limit tuning and semaphore concurrency.
- Hybrid RAG pipeline: 6 targeted queries → parallel BM25 + vector search → RRF fusion → distillation, streamed via WebSocket with Redis-backed session state and Langfuse observability.
- ReviewAgent uses 4 free LLM providers with circuit breakers and proactive fallback switching before rate limits hit; engineered 4-session concurrency using token-budget modeling against free-tier TPM/RPM/RPD limits. Deployed the project on DigitalOcean.

SYNAPSE — AI Knowledge Graph & Reasoning Engine [GitHub](#)

Python · Neo4j · Qdrant · LangGraph · LlamaIndex · FastAPI · asyncio · RAGAS · spaCy · BM25 · CrewAI · DynamoDB · GitHub Actions · React · Groq API

- Built a self-updating intelligence system that answers complex AI research questions grounded in a live knowledge graph — tracks 100K+ entities across 9 daily sources (arXiv, HuggingFace, GitHub, Semantic Scholar + more), assigns provenance tiers to every claim, and scores its own answers after every run (Faithfulness 0.889, Answer Relevancy 0.875 on live test).
- Designed a 7-node reasoning pipeline (LangGraph) that decomposes questions, retrieves from the graph, runs a 3-agent analysis crew, and self-critiques before output — completed a complex comparative research query end-to-end in 67.9s with automatic retry and escalation on failure.
- Solved the hardest free-tier constraint — API overcommit — by building a Budget Oracle node that tracks every model's rate limit in real-time and holds back requests before hitting them, not after; zero cost to run across Neo4j, Qdrant, DynamoDB, SQS, and GitHub Actions.

ACARE — Autonomous Clinical Assistance Robot | Institutionally Funded · Software Lead

ROS2 · Python · C++ · YOLOv11 · LangGraph · LLM Orchestration · Deepgram · MediaPipe · SQLite · Groq API

- Built the complete software layer for a 6-DOF clinical robot designed to fetch and deliver surgical instruments on voice command — consisted of 10 ROS2 nodes, MultiThreadedExecutor, QoS-configured typed topics, and a 10-state FSM managing every robot state from idle to emergency stop).
- Designed the interaction as fully contactless — surgeons operate the robot entirely by voice since they cannot touch anything in a sterile field; pipeline runs Silero VAD → Deepgram Nova-2 streaming STT → Groq Llama3-8b intent parsing → LangGraph dialogue manager → biometric auth gate → motion planner, with dual biometric auth (speaker verification at login + per-command voice consistency check) to prevent unauthorised commands mid-surgery.
- Added a 3-check handover before the gripper ever releases a tool — continuous face verification every 0.5s + open palm detection + voice confirmation, all three must pass simultaneously; software and firmware enforce safety independently so no single failure can cause harm; always-on ESTOP thread responds to voice in under 200ms regardless of what else the system is doing.
- Reduced instrument search time across sessions by maintaining a per-zone Bayesian probability map that updates on every find or miss, clamped to prevent runaway certainty, and persists to disk on shutdown so the robot gets smarter across uses; instrument detection runs on a self-trained YOLOv11 model (~96% mAP) on a self-curated surgical instrument dataset.

SKILLS

Languages: Python, JavaScript, C / C++, Java

AI Engineering: LangGraph, LlamaIndex, CrewAI, Multi-Agent Orchestration, RAG (BM25 + Vector + RRF), RAGAS, Prompt Engineering, YOLOv11 TFLite INT8, spaCy, MediaPipe, Groq, Gemini API, Langfuse, OpenAI API

Backend / Infra: FastAPI, asyncio, WebSocket, Redis, SQLite+FTS5+sqlite-vec, Neo4j, Qdrant, PostgreSQL, Docker, GitHub Actions, AWS (EC2, DynamoDB, SQS, S3, CloudFront), React + Vite, React Native, Ubuntu

EDUCATION / CERTIFICATIONS

Ramaiah Institute of Technology, Bengaluru

2022 – 2026

B.E. Electronics & Communication Engineering | CGPA: 7.47 / 10

Relevant Coursework: Data Structures, Operating Systems, Computer Networks, Python, Java, C/C++

Machine Learning Specialization — Coursera

2026